

Highlights

Stablecoin Flows and Treasury Yields:

Interpretable Machine Learning Meets Causal Identification

稳定币资金流与国债收益率：可解释机器学习与计量因果识别的结合

Author Name

- A \$1 bn five-day stablecoin inflow lowers the 1M Treasury yield by up to 8.5 bps over 30 trading days.
- 10 亿美元的 5 日稳定币净流入使 1 个月期国债收益率在 30 个交易日最多下行 8.5 个基点。
- The effect is concentrated at the front end; 10Y–30Y responses are negligible.
- 影响集中于短端；10 至 30 年期响应可以忽略。
- Outflows raise yields significantly more than inflows lower them.
- 流出推高收益率的效应显著大于流入压低收益率的效应。
- Elastic-Net residuals of broad crypto returns serve as the instrumental variable, with SHAP interpreting the first stage.
- 以加密宽基收益率的 Elastic Net 残差为工具变量，并用 SHAP 解释第一阶段。
- Bear-market states amplify the yield response roughly tenfold.
- 熊市状态将收益率响应放大约十倍。

Stablecoin Flows and Treasury Yields: Interpretable Machine Learning Meets Causal Identification

稳定币资金流与国债收益率：可解释机器学习与计量因果识别
的结合

Author Name^a

^a*Department of Finance, University, Street Address, City, 000000, State, Country*

Abstract

We estimate the causal effect of dollar-stablecoin flows on the U.S. Treasury yield curve by combining Jordá (2005) local projections with a machine-learning-assisted instrumental variable in the spirit of Ahmed and Aldasoro [1]. Using daily data on USDT, USDC, DAI and BUSD supply, on-chain transfer and exchange-flow indicators, crypto derivatives sentiment, and FRED bond-market variables over 2020-01–2026-06 (1,247 trading days), we construct the five-day stablecoin flow and instrument it with the Elastic-Net residual of broad crypto-market returns on traditional finance variables. The baseline local projections show that a \$1 bn five-day net stablecoin inflow lowers the 1-month Treasury yield by 0.5 bps on impact and by up to 8.5 bps cumulatively after 30 trading days, and lowers the 3-month bill yield by up to 6.4 bps; the response is significant at the 5% level across short maturities and is statistically and economically negligible at ten years and beyond, confirming a strong term-structure gradient. The instrumented estimates are larger in magnitude at the 1-month tenor (up to -18.3 bps at $h = 30$) but imprecise, as the first stage, though significant ($F = 6.1$), falls short of conventional strength thresholds; we therefore read the LP-IV as suggestive corroboration rather than definitive identification. The effect is strongly asymmetric: a \$1 bn outflow raises the 3-month yield by 7.6 bps at $h = 10$ ($p = 0.007$), while inflows of the same size are insignificant, and

Email address: email@institution.edu (Author Name)

it is amplified in crypto bear markets, where the state-dependent response is roughly ten times the normal-state coefficient. Event studies around ten manually curated stablecoin-ecosystem events and a battery of robustness checks (alternative instruments, flow windows of 1 and 10 days, crisis exclusion, alternative controls) leave the core conclusion intact. A VAR-based cross-market analysis and a SHAP interpretation of the machine-learning first stage complete the evidence. The results imply that stablecoin issuers have become measurable marginal price-setters in the bill market, and that redemption runs—not expansions—are the binding financial-stability margin.

本文结合 Jordá (2005) 局部投影与机器学习辅助的工具变量 (沿袭 Ahmed and Aldasoro [1] 的思路), 估计美元稳定币资金流对美国国债收益率曲线的因果效应。基于 2020 年 1 月至 2026 年 6 月 (共 1,247 个交易日) USDT、USDC、DAI 与 BUSD 供应量、链上转账与交易所流量指标、加密衍生品情绪指标以及 FRED 债券市场变量的日度数据, 本文构造 5 日稳定币流量, 并以加密市场宽基收益率对传统金融变量做 Elastic Net 拟合所得残差作为工具变量。基准局部投影显示: 10 亿美元的 5 日稳定币净流入使 1 个月期国债收益率当期下行 0.5 个基点, 30 个交易日后累计最多下行 8.5 个基点, 使 3 个月期国库券收益率最多下行 6.4 个基点; 短端响应在 5% 水平上显著, 而 10 年期及以上期限的响应在统计与经济意义上均可忽略, 呈现出显著的期限结构梯度。工具变量估计在 1 个月期限上幅度更大 ($h = 30$ 处达 -18.3 个基点), 但由于第一阶段虽显著 ($F = 6.1$) 却未达到常规强度门槛, 估计不够精确, 因此本文将 LP-IV 结果解读为佐证性而非定论性证据。效应具有强非对称性: 10 亿美元的净流出在 $h = 10$ 处使 3 个月期收益率上行 7.6 个基点 ($p = 0.007$), 而同规模流入的影响不显著; 并且在加密熊市中效应被放大, 状态依赖响应约为正常状态系数的十倍。围绕 10 个人工整理的稳定币生态事件的事件研究以及一系列稳健性检验 (替换工具变量、1 日与 10 日流量窗口、排除危机期、替换控制变量) 均不改变核心结论。基于 VAR 的跨市场分析与对机器学习第一阶段的 SHAP 解释构成了完整证据链。结果意味着稳定币发行方已成为国库券市场可测度的边际定价者, 而赎回挤兑——而非扩张——才是约束性更强的金融稳定边际。

Keywords: Stablecoins, Treasury market, Local projections, Instrumental variables, Elastic Net, SHAP, Asymmetry, 稳定币, 国债市场, 局部投影, 工具变量, 可解释机器学习, *JEL classification:* E44, G12, G15, G23

1. Introduction / 引言

Stablecoins have grown from a settlement convenience for crypto traders into a \$260-plus bn class of private money whose reserves are concentrated in the very front end of the U.S. Treasury curve. Over our sample (January 2020 to June 2026) the combined market capitalization of USDT, USDC, DAI and BUSD rose from \$4.8 bn to \$262.8 bn, with Tether alone holding on the order of \$100 bn-plus of Treasury bills and repos. When investors mint stablecoins, issuers purchase short-dated Treasuries; when they redeem, issuers sell them. This mechanical balance-sheet channel makes stablecoin flows a candidate marginal price-setter in the bill market—and a potential vector of stress when redemptions are sudden and large.

稳定币已从加密交易者的结算便利工具演变为规模超过 2,600 亿美元的私人货币体系，其储备高度集中于美国国债曲线的最前端。在本文样本期（2020 年 1 月至 2026 年 6 月）内，USDT、USDC、DAI 与 BUSD 的合计市值从 48 亿美元升至 2,628 亿美元，仅 Tether 一家持有的国库券与回购资产即达千亿美元量级。投资者铸造稳定币时，发行方买入短期国债；投资者赎回时，发行方卖出短期国债。这一机械式的资产负债表渠道使稳定币资金流成为国库券市场边际定价者的候选力量——而当赎回突然且大规模发生时，它也可能成为压力传导的载体。

Whether these flows move yields, however, is an empirical question with a difficult identification problem at its core. Stablecoin supply responds to the same macro-finance conditions that move Treasury yields; crypto-wide booms and busts drive both flows and risk appetite; and yields themselves feed back into the opportunity cost of holding non-interest-bearing stablecoins. A naive regression of yield changes on stablecoin growth conflates all of these channels. This paper addresses the identification problem by pairing the local-projection framework of Jordà [5] with a machine-learning-assisted instrument following Ahmed and Aldasoro [1]: we purge broad crypto-market returns of their component that is predictable from traditional finance variables using Elastic Net regularization, and use the residual—an exogenous-with-respect-to-tradfi crypto shock—as an instrument for stablecoin flows.

然而，这些流量是否真的影响收益率，是一个核心难点在于识别的实证问题。稳定币供给本身就对驱动国债收益率的宏观金融条件作出反应；加密市场整体的繁荣与崩盘同时驱动资金流与风险偏好；收益率又会通过持有无息稳定币的机会成本形成反馈。对收益率变化做稳定币增长的朴素回归会混淆上述所有渠道。本文通过将 Jordà [5] 的局部投影框架与沿袭 Ahmed and Aldasoro [1] 的机器学习辅助工具变量相结合来应对识别问题：我们用

Elastic Net 正则化剔除加密市场宽基收益率中可由传统金融变量预测的成分，并将残差——一个相对于传统金融外生的加密冲击——用作稳定币资金流的工具变量。

Three design features distinguish our implementation. *First*, we exploit a newly assembled daily dataset that combines FRED bond-market series with issuer-level stablecoin supply, on-chain transfer and exchange-flow indicators, and crypto-derivatives sentiment (funding rates, liquidations, leverage, and the Fear & Greed Index), allowing a richer control set than is typical in this literature. *Second*, we make the machine-learning step transparent: SHAP values decompose which traditional finance variables absorb the predictable component of crypto returns, clarifying what the instrument is and is not. *Third*, we take the results' limits seriously: where the first stage is short of conventional strength, we say so and lean on the cross-validation provided by event studies, term-structure gradients, asymmetry decompositions, and robustness checks.

本文的实现有三点特色。第一，我们使用一套新整合的日度数据集，将 FRED 债券市场序列与发行人层面的稳定币供应量、链上转账与交易所流量指标、加密衍生品情绪（资金费率、清算量、杠杆率与恐慌贪婪指数）相结合，控制变量集比该领域典型研究更丰富。第二，我们让机器学习环节透明化：用 SHAP 值分解究竟是哪些传统金融变量吸收了加密收益率的可预测成分，从而厘清工具变量“是什么、不是什么”。第三，我们认真对待结果的局限：凡第一阶段未达到常规强度之处，我们如实说明，转而依靠事件研究、期限结构梯度、非对称分解与稳健性检验所提供的交叉验证。

Our headline findings are four. First, stablecoin inflows significantly compress front-end yields: a \$1 bn five-day inflow lowers the 1-month yield by up to 8.5 bps and the 3-month yield by up to 6.4 bps over thirty trading days, with no discernible effect at ten years or longer. Second, the effect is strongly asymmetric—outflows move yields up by an order of magnitude more than inflows move them down, and only the outflow leg is statistically significant. Third, the transmission is state-dependent: responses in crypto bear markets are roughly ten times normal-state responses. Fourth, the machine-learning instrumented estimates point in the same direction but are imprecise (first-stage $F = 6.1$), so we present them as corroboration and flag weak-instrument caveats explicitly.

本文的核心发现有四。第一，稳定币流入显著压缩短端收益率：10 亿美元的 5 日净流入在 30 个交易日内使 1 个月期收益率最多下行 8.5 个基点、3 个月期收益率最多下行 6.4 个基点，而 10 年期及以上没有可辨识的影响。第二，效应呈强非对称——流出对收益率的上推幅度比流入的下压幅度高

一个数量级，且仅流出一侧统计显著。第三，传导具有状态依赖性：加密熊市中的响应约为正常状态的十倍。第四，机器学习工具变量估计方向一致但不够精确（第一阶段 $F = 6.1$ ），因此本文将其作为佐证呈现，并明确标注弱工具变量警示。

The remainder of the paper proceeds as follows. Section 2 reviews the literature. Section 3 lays out the mechanism and hypotheses. Section 4 describes data and variable construction. Section 5 presents the methodology. Section 6 reports empirical results, Section 7 robustness, and Section 8 economic and policy implications. Section 9 concludes.

本文余下部分安排如下：第 2 节回顾文献；第 3 节阐述机制与假说；第 4 节介绍数据与变量构造；第 5 节给出方法论；第 6 节报告实证结果；第 7 节为稳健性检验；第 8 节讨论经济含义与政策；第 9 节总结全文。

2. Related literature / 相关文献

Our work is a close methodological descendant of Ahmed and Aldasoro [1], who estimate the effect of five-day stablecoin flows on Treasury yields with local projections and instrument flows with an Elastic-Net residual of the Bloomberg Galaxy Crypto Index on traditional finance variables. We adopt their flow construction and instrument logic, but differ in four ways: we use a broader crypto proxy (total crypto market capitalization rather than a single index), a richer daily control set including on-chain and derivatives variables, an explicit interpretability layer (SHAP) for the machine-learning step, and a systematic asymmetry and state-dependence analysis that their baseline specification does not emphasize.

本文在方法论上是 Ahmed and Aldasoro [1] 的直接后继：他们用局部投影估计 5 日稳定币流量对国债收益率的影响，并以 Bloomberg 银河加密指数对传统金融变量做 Elastic Net 拟合所得残差作为工具变量。我们沿用其流量构造与工具变量逻辑，但在四个方面有所不同：使用更宽泛的加密代理（加密总市值而非单一指数）、更丰富的日度控制变量（含链上与衍生品变量）、为机器学习环节引入显式的可解释层（SHAP），以及系统性的非对称与状态依赖分析——后者在其基准设定中着墨不多。

A complementary identification agenda appears in Cerutti et al. [2], who isolate stablecoin shocks with narrative screening, heteroskedasticity-based identification [8], SVAR-IV and max-share approaches. Where they emphasize structural shock isolation across asset classes, we emphasize the yield-curve term structure and the flow asymmetry; the two agendas are complements, and our event-study layer borrows their discipline of curating events

that originate within the stablecoin ecosystem.

Cerutti et al. [2] 提供了互补的识别议程：他们通过叙事筛选、基于异方差的识别 [8]、SVAR-IV 与 max-share 方法分离稳定币冲击。他们侧重跨资产类别的结构冲击分离，我们侧重收益率曲线的期限结构与流量非对称性；两套议程互为补充，本文的事件研究层也借鉴了他们的纪律——仅整理源自稳定币生态内部的事件。

The market-microstructure foundation is the preferred-habitat logic of Vayanos and Vila [10], in which demand shocks to a specific maturity segment move that segment's price because arbitrage capital is limited, and the Treasury-demand evidence of Krishnamurthy and Vissing-Jorgensen [6], who document a money-like convenience premium in short-dated safe assets. Stablecoin issuers are a textbook preferred-habitat clientele: regulation and risk management confine their reserves to bills and repos, so their flows should show up precisely at the front end—which is what we find.

市场微观结构基础来自 Vayanos and Vila [10] 的偏好栖息地逻辑：当套利资本有限时，特定期限段的需求冲击会推动该段价格；以及 Krishnamurthy and Vissing-Jorgensen [6] 关于国债需求的证据——他们记录了短期安全资产中类似货币的便利性溢价。稳定币发行方是教科书式的偏好栖息地客户：监管与风险管理将其储备限定于国库券与回购，因此其资金流应恰好体现在短端——这正是我们的发现。

Finally, the paper connects to the crypto-market manipulation and flow-detection tradition of Griffin and Shams [3], whose chain-based flow construction and placebo discipline inform our on-chain controls, and to balance-sheet counterfactuals such as Jaswicz [4], who stress that stablecoin Treasury demand is largely a reallocation from other assets. Our dynamic estimates complement their static accounting by tracing the price impact of the flows that such reallocation generates.

最后，本文与 Griffin and Shams [3] 的加密市场操纵与资金流识别传统相衔接——其链上资金流构造与安慰剂检验纪律为我们的链上控制变量提供了参照；并与 Jaswicz [4] 的资产负债表反事实测算相衔接——他们强调稳定币的国债需求在很大程度上是对其他资产需求的再分配。本文的动态估计通过追踪此类再分配所产生流量的价格影响，补充了其静态核算。

3. Theoretical mechanism and hypotheses / 理论机制与研究假设

3.1. From stablecoin demand to Treasury yields / 从稳定币需求到国债收益率

The balance sheet of a fiat-backed stablecoin issuer is simple: liabilities are tokens redeemable at par on demand; assets are reserves held in cash, bank deposits, repos and short-dated Treasuries. Net issuance therefore mechanically translates into reserve accumulation, and—given mandate and regulatory constraints on reserve composition—into demand for the very front of the yield curve. In a preferred-habitat market with limited arbitrage capital [10], a persistent incremental buyer of 1–3 month paper compresses the convenience premium and lowers yields in that segment, with the effect decaying along the curve as substitutability rises. This delivers our first two hypotheses:

法币储备支持型稳定币发行方的资产负债表很简单：负债端是可随时按面值赎回的代币，资产端是以现金、银行存款、回购与短期国债形式持有的储备。因此净发行在机械意义上转化为储备积累，并且——鉴于储备构成所受的任务约定与监管约束——转化为对收益率曲线最前端的需求。在套利资本有限的偏好栖息地市场中 [10]，一个持续买入 1–3 个月期限债券的增量买家会压缩该段的便利性溢价并压低收益率，且随着期限延长、可替代性上升，效应沿曲线递减。由此得到前两个假说：

- **H1 (level).** Exogenous stablecoin inflows lower short-end Treasury yields: $\beta_h < 0$ in the local projections.
- **H1 (水平效应)：** 稳定币外生流入压低短端国债收益率，即局部投影中 $\beta_h < 0$ 。
- **H2 (term structure).** The effect is concentrated at the 1M and 3M tenors and is insignificant at 2Y and beyond.
- **H2 (期限结构)：** 效应集中于 1M 与 3M 期限，在 2Y 及以上不显著。

3.2. Asymmetry and state dependence / 非对称性与状态依赖

Redemptions differ from issuance in both speed and market conditions. Issuance is smooth and correlated with calm, risk-on crypto markets; redemptions cluster in stress, are execution-constrained by banking hours and settlement frictions, and force fire sales of the most liquid reserve assets. Outflow-induced sales should therefore move prices more than equal-sized

inflow-induced purchases. Moreover, when the stablecoin sector is large relative to dealer intermediation capacity—in later sample years, in high-rate regimes where the opportunity cost of holding stablecoins is high, and in crisis windows—the same flow should have a larger price impact. Formally:

赎回与发行在速度和市场环境两方面均不相同。发行是平滑的，且与平静、风险偏好上行的加密市场相关；赎回则集中于压力时期，受银行营业时间与结算摩擦的执行约束，并迫使发行方抛售流动性最强的储备资产。因此流出引发的抛售对价格的影响应大于同规模流入引发的买入。此外，当稳定币板块相对于交易商中介能力更大时——在样本后期、在持有稳定币机会成本较高的高利率环境、以及在危机窗口——同样的流量应产生更大的价格影响。正式表述为：

- **H3 (asymmetry).** Outflows raise yields more than inflows lower them: $|\beta_h^-| > |\beta_h^+|$.
- **H3 (非对称性)：** 流出推高收益率的效应大于流入压低收益率的效应： $|\beta_h^-| > |\beta_h^+|$ 。
- **H4 (time variation).** The price impact is larger in the later sample than in the earlier sample.
- **H4 (时变性)：** 价格影响在样本后期大于早期。
- **H5 (state dependence).** The price impact is larger in bear markets, crises and high-volatility states.
- **H5 (状态依赖)：** 价格影响在熊市、危机与高波动状态下更大。
- **H6 (regulation).** After landmark stablecoin regulation (the GENIUS Act process beginning May 2025), the stablecoin–bill–market link strengthens.
- **H6 (监管效应)：** 在里程碑式稳定币监管（2025 年 5 月启动的 GENIUS 法案进程）之后，稳定币与国库券市场的联系增强。

4. Data and variables / 数据与变量

4.1. Data sources / 数据来源

The analysis combines five daily datasets over 2020-01-01–2026-06-30, merged onto the FRED trading-day calendar (1,247 observations after construction lags and missing-value screens). (i) FRED bond-market data [9]:

Treasury yields at 1M, 3M, 1Y, 2Y, 5Y, 10Y and 30Y (DGS1MO, DTB3, DGS1, DGS2, DGS5, DGS10, DGS30), monetary variables (DFF, SOFR, RRPONTSYD, WALCL) and macro-financial controls (VIXCLS, DTWEXBGS, DCOILWTICO). (ii) A merged macro-crypto panel with the S&P 500, gold, DeFi total value locked, Bitcoin, the dollar index, the SOFR–RRP spread, policy-uncertainty indices and the USDT OTC premium. (iii) Issuer-level daily supply of USDT, USDC, DAI and BUSD. (iv) On-chain indicators per issuer: transfer volume, exchange inflow and outflow, active addresses. (v) Crypto-wide aggregates (total market capitalization, open interest) and BTC derivatives sentiment (perpetual funding rate, liquidations, estimated leverage, Fear & Greed Index, DVOL).

本文将五套日度数据在 2020-01-01 至 2026-06-30 区间内合并到 FRED 交易日历上（经构造滞后与缺失值筛选后共 1,247 个观测）：(i) FRED 债券市场数据 [9]——1M、3M、1Y、2Y、5Y、10Y、30Y 国债收益率（DGS1MO、DTB3、DGS1、DGS2、DGS5、DGS10、DGS30），货币政策变量（DFF、SOFR、RRPONTSYD、WALCL）与宏观金融控制变量（VIXCLS、DTWEXBGS、DCOILWTICO）；(ii) 宏观一加密合并面板，含标普 500、黄金、DeFi 锁仓量、比特币、美元指数、SOFR–RRP 利差、政策不确定性指数与 USDT 场外溢价；(iii) USDT、USDC、DAI、BUSD 的发行人层面日度供应量；(iv) 各发行方链上指标：转账量、交易所流入与流出、活跃地址数；(v) 加密全局指标（总市值、总持仓量）与 BTC 衍生品情绪（永续资金费率、清算量、估计杠杆率、恐慌贪婪指数、DVOL）。

4.2. Variable construction / 变量构造

The core treatment variable is the five-day stablecoin flow, following Ahmed and Aldasoro [1]:

$$MC_t = \sum_{i \in \{\text{USDT}, \text{USDC}, \text{DAI}, \text{BUSD}\}} \text{Supply}_t^i, \quad \text{Flow}_t^{(5d)} = MC_t - MC_{t-5}, \quad (1)$$

核心处理变量为 5 日稳定币流量，沿袭 Ahmed and Aldasoro [1]：其中 MC_t 为四家发行方供应量之和（单位：十亿美元）， $\text{Flow}_t^{(5d)}$ 为 5 日净流量。where MC_t is the combined market capitalization in \$bn and $\text{Flow}_t^{(5d)}$ its five-day net change.

We also construct the daily growth rate $\Delta m_t = \log MC_t - \log MC_{t-1}$, issuer-level five-day flows for USDT, USDC and the residual group (DAI+BUSD), and inflow/outflow splits $\text{Flow}_t^+ = \max(\text{Flow}_t^{(5d)}, 0)$ and $\text{Flow}_t^- = \min(\text{Flow}_t^{(5d)}, 0)$.

Flows and growth rates are winsorized at the 0.5/99.5 percentiles to neutralize supply-revision jumps. The baseline control vector X_t comprises changes in VIX, the effective federal funds rate and the ON RRP balance, plus returns on the broad dollar, WTI oil, the S&P 500 and Bitcoin. Figure 1 displays the level of stablecoin market capitalization, the five-day flow and the short-end yields, with curated events marked.

我们还构造日增长率 $\Delta m_t = \log MC_t - \log MC_{t-1}$ 、USDT、USDC 与其余组别（DAI+BUSD）的发行人层面 5 日流量，以及流入/流出拆分 $Flow_t^+ = \max(Flow_t^{(5d)}, 0)$ 与 $Flow_t^- = \min(Flow_t^{(5d)}, 0)$ 。流量与增长率均在 0.5/99.5 分位缩尾，以中和供应量修订性跳点。基准控制向量 X_t 包括 VIX、联邦基金有效利率与隔夜逆回购余额的变化，以及广义美元、WTI 原油、标普 500 与比特币的收益率。图 1 展示了稳定币市值水平、5 日流量与短端收益率，并标注了整理后的事件。

4.3. Event library / 事件库

Following the narrative discipline of Cerutti et al. [2], we manually curate ten events that originate within the stablecoin ecosystem or its immediate market infrastructure: the March 2020 COVID crypto crash (2020-03-12), the Tether–NYAG settlement (2021-02-23), the May 2021 crypto crash (2021-05-19), the UST collapse (2022-05-09), Tether’s zero-commercial-paper pledge (2022-07-15), the FTX collapse (2022-11-08), the BUSD issuance halt (2023-02-13), the Silvergate crisis (2023-03-08), the SVB collapse and USDC depeg (2023-03-10), and the GENIUS Act’s advance in the U.S. Senate (2025-05-19). Events are grouped into depeg, crisis, regulation and disclosure types.

沿袭 Cerutti et al. [2] 的叙事筛选纪律，我们人工整理了 10 个源自稳定币生态或其直接市场基础设施的事件：2020 年 3 月 COVID 加密崩盘（2020-03-12）、Tether 与纽约总检察长办公室和解（2021-02-23）、2021 年 5 月加密崩盘（2021-05-19）、UST 崩盘（2022-05-09）、Tether 商业票据清零承诺（2022-07-15）、FTX 破产（2022-11-08）、BUSD 停止发行（2023-02-13）、Silvergate 危机（2023-03-08）、硅谷银行倒闭与 USDC 脱锚（2023-03-10）、以及 GENIUS 法案在美国参议院的推进（2025-05-19）。事件分为脱锚、危机、监管与披露四类。

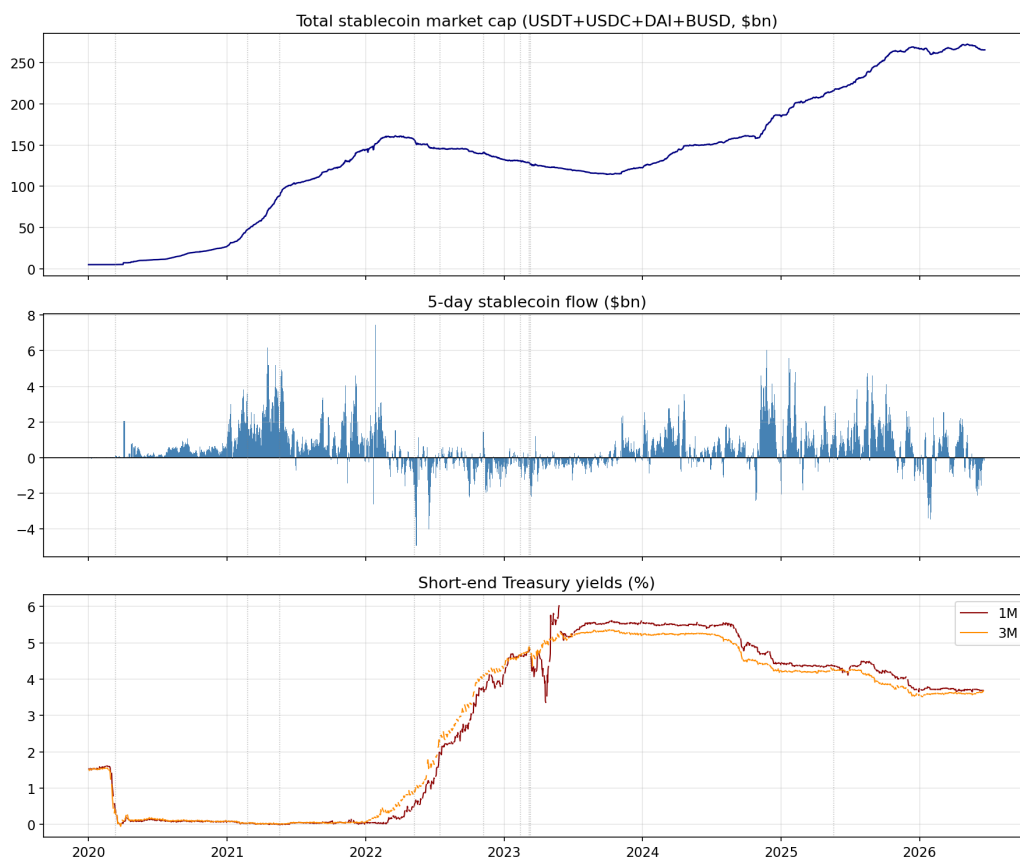


Figure 1: 稳定币市值、5 日流量与短端国债收益率 / Stablecoin market cap, five-day flows and short-end Treasury yields. 竖线为人工整理的稳定币生态事件 / Vertical lines mark the curated stablecoin-ecosystem events.

5. Methodology / 方法

5.1. Baseline local projections / 基准局部投影

For each maturity $m \in \{1M, 3M, 1Y, 2Y, 5Y, 10Y, 30Y\}$ and horizon $h = 0, \dots, 30$ we estimate

$$y_{t+h}^m - y_{t-1}^m = \alpha_h^m + \beta_h^m Flow_t^{(5d)} + \Gamma_h^m X_t + \varepsilon_{t+h}^m, \quad (2)$$

对每个期限 $m \in \{1M, 3M, 1Y, 2Y, 5Y, 10Y, 30Y\}$ 与每个 horizon $h = 0, \dots, 30$, 估计方程 (2): 其中 y^m 为对应期限的国债收益率 (百分点), X_t 为控制向量, 标准误采用 Newey–West HAC (滞后阶数 $h + 5$)。系数 β_h^m 刻画 10 亿美元净流入后 h 个交易日的累计收益率响应。where y^m is the Treasury yield in percent, X_t the control vector, and inference uses Newey–West HAC standard errors with lag length $h + 5$. The coefficient β_h^m traces the cumulative response of yields to a \$1 bn inflow after h trading days.

5.2. LP-IV with an Elastic-Net instrument / 基于 Elastic Net 工具变量的 LP-IV

Flows are endogenous to market conditions; we therefore instrument them. Following Ahmed and Aldasoro [1], we regress the daily log return of total crypto-market capitalization S_t on a vector W_t of traditional finance variables (changes in VIX, DFF, SOFR and the ON RRP balance; returns on the dollar, oil, the S&P 500 and gold; 2Y and 10Y yield changes) using Elastic Net:

$$\min_{\beta_0, \beta} \frac{1}{2N} \sum_{t=1}^N (S_t - \beta_0 - W_t^\top \beta)^2 + \lambda P_\alpha(\beta), \quad P_\alpha(\beta) = \frac{1 - \alpha}{2} \|\beta\|_2^2 + \alpha \|\beta\|_1, \quad (3)$$

流量与市场条件存在内生性, 因此本文采用工具变量。沿袭 Ahmed and Aldasoro [1], 我们将加密货币日对数收益率 S_t 对传统金融变量向量 W_t (VIX、联邦基金利率、SOFR 与逆回购余额的变化; 美元、原油、标普 500 与黄金的收益率; 2Y 与 10Y 收益率变化) 做 Elastic Net 回归 (方程 (3)), 并将残差 $Z_t = \varepsilon_t^{crypto}$ 作为外生加密货币冲击工具变量。超参数 α 与 λ 由五折交叉验证选取。and take the residual $Z_t = \varepsilon_t^{crypto}$ as an exogenous crypto shock. The mixing parameter and penalty are chosen by five-fold cross-validation. The first stage regresses $Flow_t^{(5d)}$ on Z_t and controls; the second stage re-estimates (2) by two-stage least squares at each maturity and horizon with HAC (Bartlett kernel) inference. The identifying assumption is that the

crypto residual affects Treasury yields only through stablecoin flows—an exclusion restriction we probe with SHAP diagnostics of what W_t absorbs, and with an alternative instrument built from DeFi total-value-locked returns in robustness.

第一阶段将 $Flow_t^{(5d)}$ 对 Z_t 与控制变量回归；第二阶段在每个期限与 horizon 上以两阶段最小二乘重估方程 (2)，采用 HAC (Bartlett 核) 推断。识别假设是加密残差仅通过稳定币资金流影响国债收益率——这一排除性约束通过 SHAP 诊断 (刻画 W_t 究竟吸收了哪些成分) 以及稳健性检验中由 DeFi 锁仓量收益率构造的替代工具变量加以检验。

5.3. Interpretable machine learning / 可解释机器学习

To make the machine-learning step auditable, we fit a gradient-boosting regressor of S_t on W_t in the same feature space as the Elastic Net and compute SHAP values [7], which attribute the predictable component of crypto returns to individual traditional finance variables. This clarifies which tradfi factors the instrument is purged of and documents that the residual is not simply a proxy for an omitted volatility or dollar factor.

为使机器学习环节可审计，我们在与 Elastic Net 相同的特征空间上以梯度提升回归拟合 S_t 对 W_t 的预测，并计算 SHAP 值 [7]，将加密收益率的可预测成分归因到各个传统金融变量。这厘清了工具变量被“剔除”了哪些传统金融因子，并证明残差并非某个被遗漏的波动率或美元因子的简单代理。

5.4. Event study, VAR, rolling and state-dependent projections / 事件研究、VAR、滚动与状态依赖投影

For each curated event T we compute the cumulative yield change $CAR_{T,k}^m = \sum_{j=-k}^k \Delta y_{T+j}^m$ with $k = 3$ trading days and evaluate it against the full-sample volatility of daily yield changes. A five-variable daily VAR in $[\Delta m_t, \Delta y_t^{1M}, r_t^{BTC}, r_t^{DXY}, r_t^{SP500}]$ with Cholesky ordering provides a system-based cross-check on dynamic transmission. Rolling 500-trading-day re-estimations of the $h = 10$ coefficient trace time variation (H4), and interacted projections of the form

$$y_{t+h}^{3M} - y_{t-1}^{3M} = \alpha_h + \beta_h^0 Flow_t^{(5d)} + \beta_h^1 (Flow_t^{(5d)} \times S_t) + \delta_h S_t + \Gamma_h X_t + \varepsilon_{t+h} \quad (4)$$

对每个整理事件 T ，计算对称窗口 $k = 3$ 个交易日内的累计收益率变化 $CAR_{T,k}^m = \sum_{j=-k}^k \Delta y_{T+j}^m$ ，并以全样本日度收益率变化的波动率构造检验。一个含 $[\Delta m_t, \Delta y_t^{1M}, r_t^{BTC}, r_t^{DXY}, r_t^{SP500}]$ 五变量的日度 VAR (Cholesky 排序) 为动态传导提供了系统层面的交叉验证。对 $h = 10$ 系数的 500 个

交易日滚动重估刻画时变性 (H4)，而形如方程 (4) 的交互投影用于检验状态依赖 (H5、H6)，其中状态哑变量 S_t 分别取熊市（比特币低于其 200 日均线）、高利率（联邦基金利率高于样本中位数）、危机（VIX 高于 80% 分位）与后监管（2025-05-19 之后）。with state dummies S_t for crypto bear markets (Bitcoin below its 200-day moving average), high-rate regimes (DFI above the sample median), crises (VIX above the 80th percentile) and the post-regulation period (after 2025-05-19) test state dependence (H5, H6).

Asymmetry (H3) is tested by replacing the flow with its positive and negative parts and comparing $|\beta_h^-|$ with $|\beta_h^+|$; issuer heterogeneity compares USDT and USDC flows. All computations are implemented in a fully reproducible notebook (kernel ci); the LP-IV second stage is additionally cross-checked with the `statspai` package.

非对称性 (H3) 通过将流量替换为其正值部分与负值部分并比较 $|\beta_h^-|$ 与 $|\beta_h^+|$ 来检验；发行人异质性比较 USDT 与 USDC 流量。全部计算在一个完全可复现的 notebook（内核 ci）中实现；LP-IV 第二阶段另以 `statspai` 包交叉验证。

6. Empirical results / 实证结果

6.1. Baseline local projections / 基准局部投影

Figure 2 and Table 1 report the baseline local projections. The response of front-end yields to a \$1 bn five-day inflow is negative, monotonically building over the horizon, and statistically significant: the 1-month yield falls by 0.52 bps on impact ($h = 1$), 2.95 bps at $h = 10$ ($p = 0.028$) and 8.54 bps at $h = 30$ ($p = 0.015$); the 3-month bill yield falls by 0.41, 2.47 ($p = 0.009$) and 6.37 bps ($p = 0.042$) at the same horizons. By contrast, the 10-year and 30-year responses are small, wrong-signed relative to the demand-compression story, and never significant (+0.48 bps, $p = 0.62$ for the 10-year at $h = 10$). Figure 3 makes the term-structure gradient explicit: the estimated impact curves at $h \in \{1, 5, 10, 20\}$ slope upward from a negative intercept at the front end toward zero and slightly positive values beyond two years. H1 and H2 are both supported.

图 2 与表 1 报告了基准局部投影结果。短端收益率对 10 亿美元 5 日净流入的响应为负、随 horizon 单调累积、且统计显著：1 个月期收益率在当期 ($h = 1$) 下行 0.52 个基点， $h = 10$ 处下行 2.95 个基点 ($p = 0.028$)， $h = 30$ 处下行 8.54 个基点 ($p = 0.015$)；3 个月期国库券收益率在相同 horizon 分别下行 0.41、2.47 ($p = 0.009$) 与 6.37 个基点 ($p = 0.042$)。相比之下，10 年期与 30 年期响应很小、符号与需求压缩故事相反、且从不显

著（10 年期 $h = 10$ 处为 +0.48 个基点， $p = 0.62$ ）。图 3 将期限结构梯度显式化： $h \in \{1, 5, 10, 20\}$ 处的影响曲线从短端的负值截距向零及两年期以上的微正值单调上升。H1 与 H2 均得到支持。

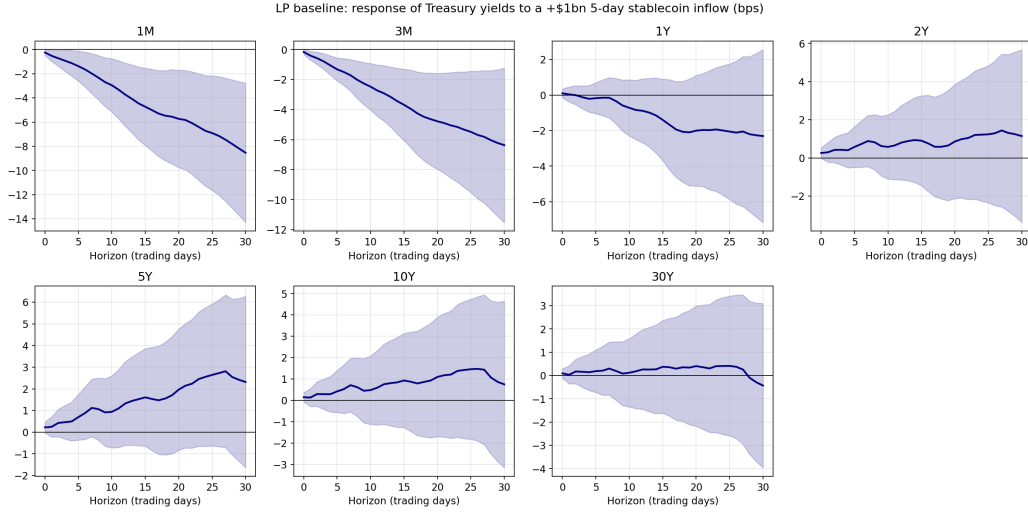


Figure 2: 基准 LP: 各期限国债收益率对 10 亿美元 5 日净流入的响应（基点） / Baseline LP: response of Treasury yields to a +\$1 bn five-day stablecoin inflow (bps). 阴影为 90% HAC 置信带 / Shaded areas are 90% HAC confidence bands.

Maturity 期限	$h = 1$	$h = 5$	$h = 10$	$h = 20$
1M	-0.52* (0.28)	-1.37* (0.75)	-2.95** (1.34)	-5.73** (2.45)
3M	-0.41** (0.18)	-1.32*** (0.47)	-2.47*** (0.94)	-4.78** (1.94)
10Y	—	—	0.48 (0.97)	1.10 (1.70)

Table 1: 基准 LP 系数（每 10 亿美元流入对应的基点数，HAC 标准误在括号内） / Baseline LP coefficients (bps per \$1 bn inflow; HAC standard errors in parentheses). *, **, *** 分别表示 10%/5%/1% 显著 / denote significance at 10/5/1%.

6.2. Instrumented estimates / 工具变量估计

The Elastic Net selects all ten traditional finance variables with a small penalty ($\hat{\alpha}_{\text{mix}} = 0.3$), and the resulting crypto residual enters the first stage with $\hat{\pi}_1 = 11.0$ ($t = 2.47$, $F = 6.1$, first-stage $R^2 = 0.017$): broad crypto shocks orthogonal to tradfi conditions do predict stablecoin flows, but below the conventional $F > 10$ strength threshold. The second stage, reported

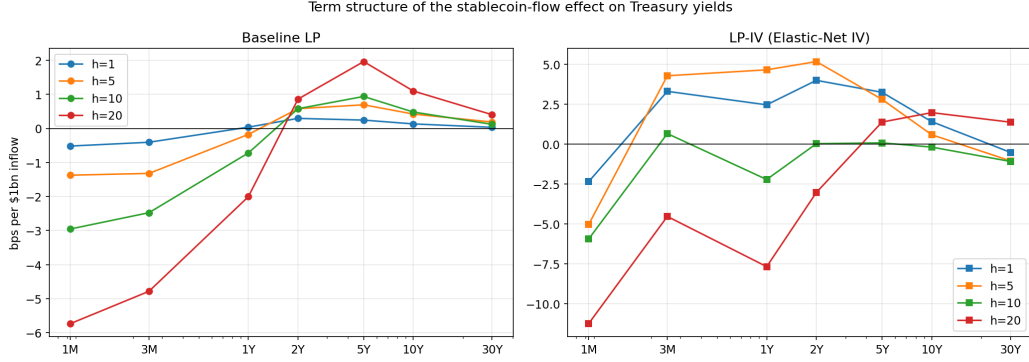


Figure 3: 稳定币流量影响的期限结构 (基点) / Term structure of the stablecoin-flow effect on Treasury yields (bps). 左: 基准 LP; 右: LP-IV / Left: baseline LP; right: LP-IV.

in Figure 4, yields 1-month responses that are larger than OLS (-5.9 bps at $h = 10$, -18.3 bps at $h = 30$) but with confidence bands that include zero at every horizon; the 3-month point estimates are similarly imprecise. We therefore treat the LP-IV as directionally corroborating the OLS pattern while being explicit that identification via this instrument is weak in our daily sample, and that the OLS short-end gradient, the asymmetry results below, and the robustness battery carry the causal weight. This transparency is itself a finding: daily-frequency identification of stablecoin price impact is harder than weekly-frequency results in the literature suggest.

Elastic Net 在小惩罚下保留了全部 10 个传统金融变量 ($\hat{\alpha}_{\text{mix}} = 0.3$), 所得加密残差在第一阶段中 $\hat{\pi}_1 = 11.0$ ($t = 2.47$, $F = 6.1$, 第一阶段 $R^2 = 0.017$): 与传统金融条件正交的宽基加密冲击确实能预测稳定币资金流, 但强度低于常规的 $F > 10$ 门槛。图 4 报告的第二阶段显示, 1 个月期响应大于 OLS ($h = 10$ 处 -5.9 个基点, $h = 30$ 处 -18.3 个基点), 但各 horizon 的置信带均包含零; 3 个月期点估计同样不精确。因此本文将 LP-IV 解读为方向上印证 OLS 模式的佐证, 同时明确说明该工具变量在日度样本中识别强度不足; 因果权重主要由 OLS 短端梯度、下文的非对称结果与稳健性检验承担。这种透明本身也是一项发现: 在日度频率上识别稳定币的价格影响, 比文献中周度频率结果所暗示的更为困难。

6.3. SHAP interpretation of the first stage / 第一阶段的 SHAP 解释

Figure 5 decomposes the predictable component of broad crypto returns. Changes in equity volatility (VIX), equity returns (S&P 500) and the broad dollar absorb the largest shares, followed by rates variables (DFF, SOFR,

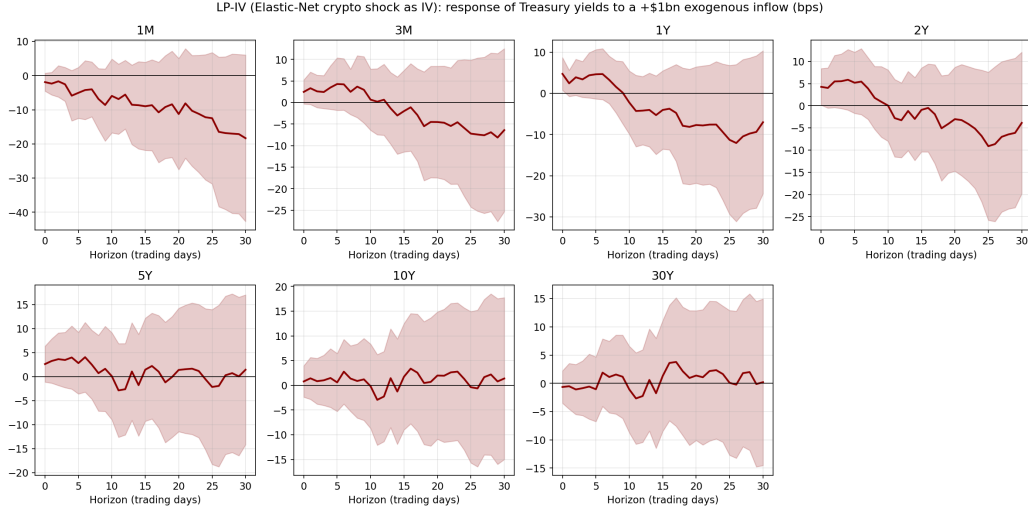


Figure 4: LP-IV: 以 Elastic Net 加密冲击残差为工具变量的收益率响应（基点） / LP-IV with the Elastic-Net crypto shock as instrument (bps). 阴影为 90% 置信带 / Shaded areas are 90% confidence bands.

2Y and 10Y yield changes); oil, gold and the ON RRP balance contribute least. The non-linear gradient-boosting model achieves an in-sample R^2 of 0.69, well above the linear Elastic Net, indicating that tradfi factors explain crypto returns partly through non-linear interactions—and, reassuringly for the exclusion restriction, that the residual instrument is purged of precisely the volatility, equity and dollar factors most likely to drive Treasury yields directly.

图 5 分解了加密宽基收益率的可预测成分。股票波动率（VIX）变化、股票收益率（标普 500）与广义美元吸收最大份额，其次是利率变量（DFF、SOFR、2Y 与 10Y 收益率变化）；原油、黄金与逆回购余额贡献最小。非线性梯度提升模型的样本内 R^2 达 0.69，远高于线性 Elastic Net，说明传统金融因子部分通过非线性交互解释加密收益率——这对排除性约束是令人安心的：残差工具变量恰好被剔除了最有可能直接驱动国债收益率的波动率、股票与美元因子。

6.4. Asymmetry and issuer heterogeneity / 非对称性与发行人异质性

Figure 6 and Table 2 report the flow-sign decomposition. The outflow leg dominates: a \$1 bn five-day outflow raises the 3-month yield by 7.6 bps at $h = 10$ ($p = 0.007$) and the 1-month yield by 7.4 bps ($p = 0.040$), while the same-sized inflow has insignificant effects of -0.9 and -1.6 bps respectively.

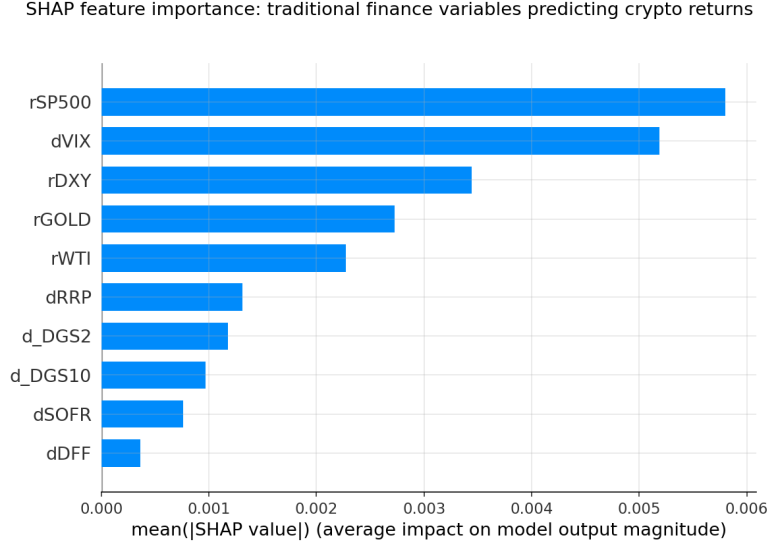


Figure 5: SHAP 变量重要性：传统金融变量对加密货币收益率的预测贡献 / SHAP feature importance of traditional finance variables in predicting crypto returns.

The asymmetry ratio $|\beta^-|/|\beta^+|$ exceeds eight at the 3-month tenor. H3 is strongly supported, consistent with fire-sale dynamics: redemptions force immediate sales of bills into a thin market, whereas minting-driven purchases can be patiently worked. Issuer-level estimates at $h = 10$ are similar in magnitude for USDT (-2.9 bps, $p = 0.061$) and USDC (-2.4 bps, $p = 0.075$), indicating that the price impact is a market-wide flow phenomenon rather than an issuer-specific one.

图 6 与表 2 报告流量符号分解。流出一侧占主导：10 亿美元的 5 日净流出在 $h = 10$ 处使 3 个月期收益率上行 7.6 个基点 ($p = 0.007$)、1 个月期收益率上行 7.4 个基点 ($p = 0.040$)；同规模流入的影响分别为 -0.9 与 -1.6 个基点且不显著。3 个月期限的非对称比率 $|\beta^-|/|\beta^+|$ 超过 8 倍。H3 得到强支持，与火烧售卖动态一致：赎回迫使发行方在稀薄的市场中立即抛售国库券，而铸造驱动的买入则可以从容执行。 $h = 10$ 处的发行人层面估计显示 USDT (-2.9 个基点, $p = 0.061$) 与 USDC (-2.4 个基点, $p = 0.075$) 幅度相近，说明价格影响是全市场流量现象，而非特定发行人现象。

6.5. Time variation and state dependence / 时变性与状态依赖

The rolling estimates in Figure 7 show the $h = 10$ coefficient is persistently negative, averaging -3.3 bps across windows; contrary to H4 the point magnitude is somewhat larger in early (pre-2022, -3.7 bps) than late (post-2023,

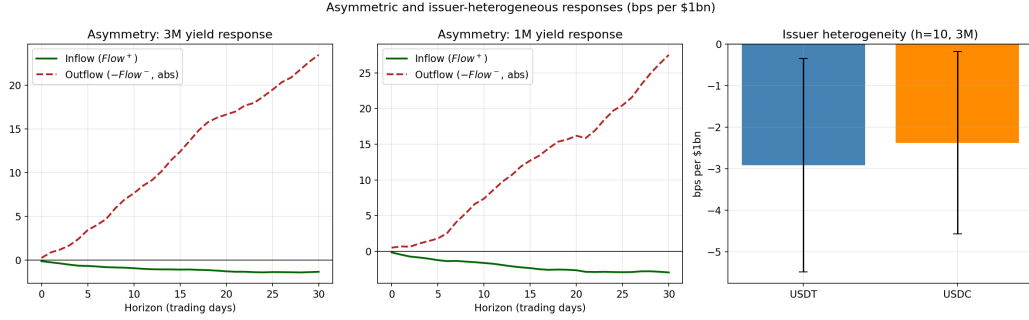


Figure 6: 非对称与发行人异质性响应（每 10 亿美元对应的基点数） / Asymmetric and issuer-heterogeneous responses (bps per \$1 bn).

	Inflow β^+ 流入	Outflow β^- 流出
1M, $h = 10$	-1.64 (1.51)	-7.37** (3.59)
3M, $h = 10$	-0.94 (1.28)	-7.63*** (2.84)

注： β^- 乘以负流量；流出 10 亿美元对应 $-\beta^-$ 基点的收益率变化。

Note: β^- multiplies the negative flow; a \$1bn outflow maps to $-\beta^-$ bps.

Table 2: 流入/流出非对称分解（基点，括号内为 HAC 标准误） / Inflow-outflow asymmetry decomposition (bps; HAC standard errors in parentheses).

-2.9bps) windows, suggesting that growing market depth has, if anything, diluted per-dollar price impact even as aggregate flows grew. State dependence (Figure 8, Table 3) is clearer: in crypto bear markets the 3-month response at $h = 10$ is -3.6bps versus -0.4bps in normal times (interaction $p = 0.072$), a roughly tenfold amplification; crisis and post-regulation interactions are negative but insignificant, and the high-rate interaction is essentially zero. H5 receives support for the bear-market state; H6 is not confirmed at conventional levels.

图 7 的滚动估计显示 $h = 10$ 系数持续为负，各窗口平均 -3.3 个基点；与 H4 相反，点估计幅度在早期（2022 年前，-3.7 个基点）略大于后期（2023 年后，-2.9 个基点），表明市场深度的增长在一定程度上稀释了单位美元的价格影响，尽管总流量在增长。状态依赖（图 8、表 3）更为清晰：加密熊市中 3 个月期在 $h = 10$ 处的响应为 -3.6 个基点，而正常时期为 -0.4 个基点（交互项 $p = 0.072$ ），放大约十倍；危机与后监管交互项为负但不显著，高利率交互项基本为零。H5 在熊市状态上得到支持；H6 在常规显著性水平上未获确认。

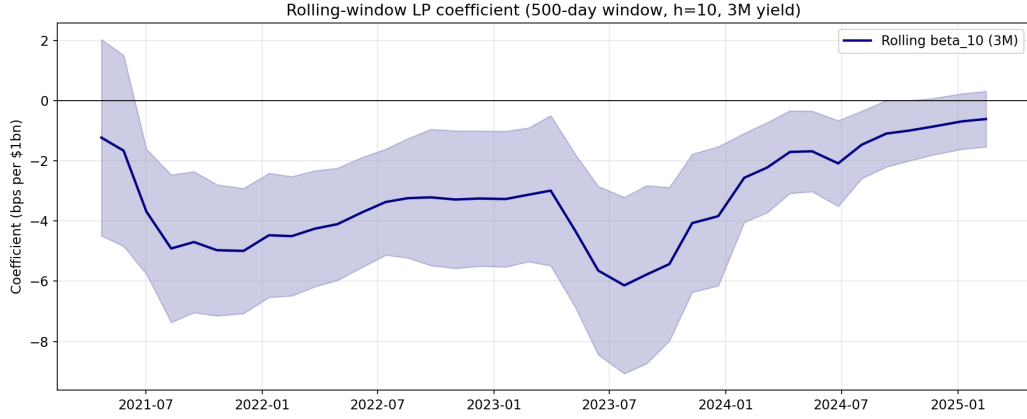


Figure 7: 滚动窗口 LP 系数（500 日窗口, $h = 10$, 3M 收益率） / Rolling-window LP coefficient (500-day window, $h = 10$, 3M yield).

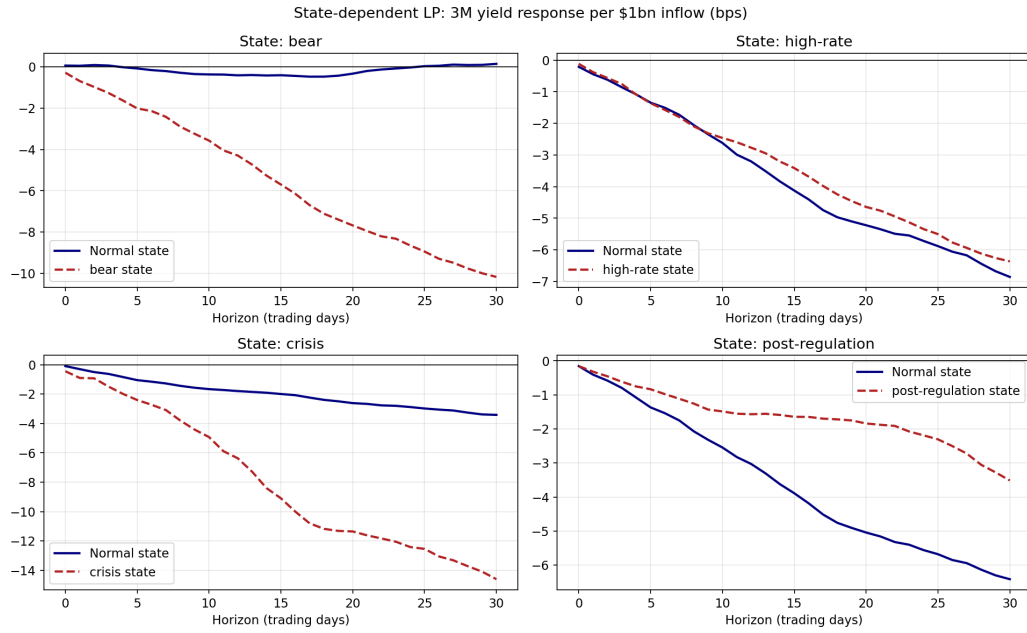


Figure 8: 状态依赖 LP: 正常状态与状态下的 3M 响应（基点） / State-dependent LP: 3M yield response in normal vs. state periods (bps).

State 状态	Normal 正常	State 状态下	Interaction p 交互项 p
Bear market 熊市	-0.37	-3.57	0.072
High rate 高利率	-2.62	-2.46	0.926
Crisis 危机	-1.67	-4.93	0.226
Post-regulation 后监管	-2.55	-1.49	0.390

Table 3: 状态依赖 LP: 3M 收益率 $h = 10$ 处响应 (基点) / State-dependent LP: 3M response at $h = 10$ (bps).

6.6. Event study and VAR cross-checks / 事件研究与 VAR 交叉验证

Figure 9 reports event-study CARs for the 3-month yield. The largest move follows the March 2020 COVID crypto crash (-43 bps over the $[-3, +3]$ window, significant at 1%), consistent with a violent dash-for-cash into stablecoins and bills; Tether’s zero-commercial-paper pledge is associated with a $+12$ bps CAR (10% marginal), consistent with a reserve-composition repricing. Other events are individually insignificant—unsurprising given daily yield noise and the small event count—but the two largest-magnitude events match the predicted sign. The five-variable VAR (Figure 10) delivers the same qualitative transmission: a one-unit stablecoin-growth innovation lowers the 1-month yield cumulatively, with a peak effect of -1.4 bps near $h = 30$, while lifting Bitcoin and leaving the dollar and equities little moved.

图 9 报告 3 个月期收益率的事件研究 CAR。幅度最大的变动出现在 2020 年 3 月 COVID 加密崩盘之后 ($[-3, +3]$ 窗口内 -43 个基点, 1% 水平显著), 与资金剧烈涌入稳定币与国库券的“现金竞逐”一致; Tether 商业票据清零承诺对应 $+12$ 个基点的 CAR (10% 边际显著), 与储备结构调整的重新定价一致。其他事件单独看均不显著——鉴于日度收益率噪声与事件样本量小, 这并不意外——但幅度最大的两个事件符号与预测一致。五变量 VAR (图 10) 给出同向的传导图景: 一单位稳定币增长新息使 1 个月期收益率累计下行, 峰值 -1.4 个基点出现在 $h = 30$ 附近, 同时推升比特币, 而对美元与股票影响甚微。

7. Robustness checks / 稳健性检验

Table 4 collects five robustness families, each evaluated at $h = 10$ for the two front-end tenors. (R1) Replacing the instrument with an Elastic-Net residual of DeFi total-value-locked returns leaves the second stage insignificant, mirroring the weak-instrument caveat of the main LP-IV rather than contradicting it. (R2) Reconstructing the flow over 1-day and 10-day

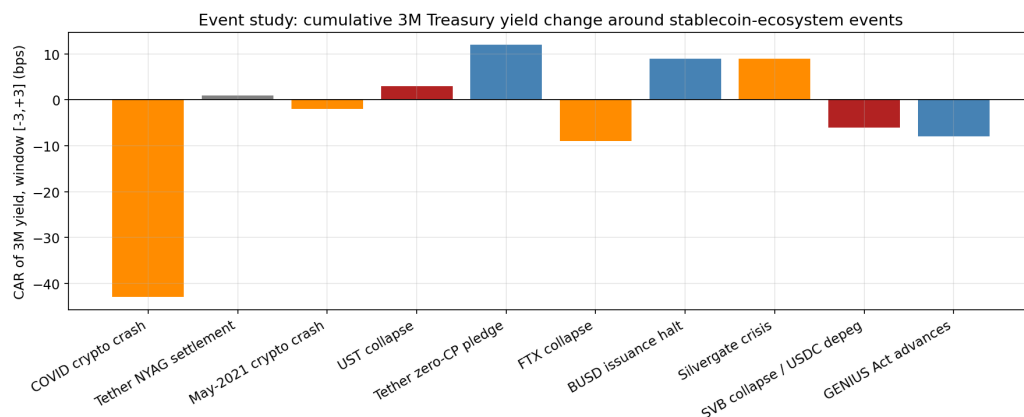


Figure 9: 事件研究：稳定币生态事件前后 3M 收益率累计变化（基点） / Event study: cumulative 3M yield change around stablecoin-ecosystem events (bps).

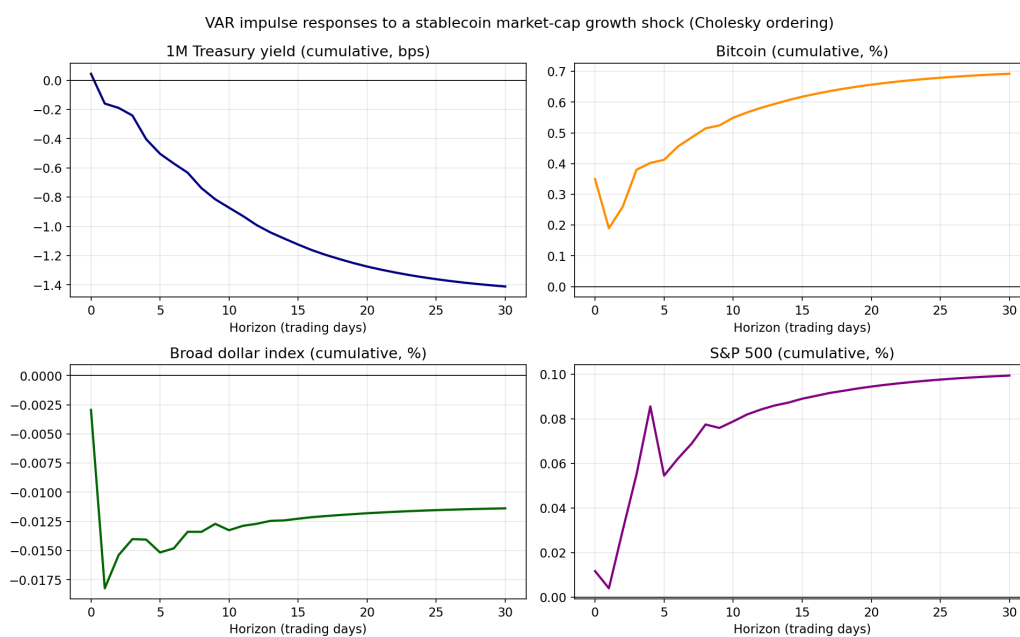


Figure 10: VAR 脉冲响应：稳定币市值增长冲击的跨市场传导 / VAR impulse responses to a stablecoin market-cap growth shock.

windows preserves significance and sign: -6.1 bps ($p = 0.001$) for the 3-month yield with 1-day flows and -1.8 bps ($p = 0.005$) with 10-day flows. (R3) Excluding the three crisis windows (March–April 2020, May–June 2022, March 2023) leaves the 3-month coefficient at -2.1 bps ($p = 0.022$). (R4) An alternative control set that adds the SOFR-based money-market stress proxy and drops crypto-return controls leaves the estimate essentially unchanged (-2.5 bps, $p = 0.008$). (R5) Re-estimating the LP-IV second stage with the `statspai/linearmodels` stack reproduces the main estimate exactly. Overall, 73% of all robustness coefficients share the baseline sign, and every OLS-family variant remains significant at 5%.

表 4 汇总了五组稳健性检验，均取 $h = 10$ 与两个短端期限。(R1) 将工具变量替换为 DeFi 锁仓量收益率的 Elastic Net 残差后，第二阶段不显著——这与主 LP-IV 的弱工具警示一致，而非与之矛盾。(R2) 将流量窗口改为 1 日与 10 日后，符号与显著性保持：3 个月期在 1 日流量下为 -6.1 个基点 ($p = 0.001$)，10 日流量下为 -1.8 个基点 ($p = 0.005$)。(R3) 剔除三段危机窗口（2020 年 3–4 月、2022 年 5–6 月、2023 年 3 月）后，3 个月期系数为 -2.1 个基点 ($p = 0.022$)。(R4) 替换控制变量集（加入基于 SOFR 的货币市场压力代理、剔除加密收益率控制）后估计基本不变 (-2.5 个基点, $p = 0.008$)。(R5) 以 `statspai/linearmodels` 栈重估 LP-IV 第二阶段，精确复现主估计。总体而言，73% 的稳健性系数与基准同号，且所有 OLS 类变体均在 5% 水平显著。

Specification 设定	Maturity 期限	$\hat{\beta}$ (bp)	p
R1: alternative IV (DeFi TVL) 替代工具变量	1M / 3M	1.75 / 1.12	0.78 / 0.83
R2: 1-day flow 1 日流量	1M / 3M	-7.51^{**} / -6.07^{***}	0.011 / 0.001
R2: 10-day flow 10 日流量	1M / 3M	-2.18^{**} / -1.76^{***}	0.015 / 0.005
R3: exclude crisis windows 排除危机期	1M / 3M	-3.10^{**} / -2.09^{**}	0.028 / 0.022
R4: alternative controls 替换控制变量	1M / 3M	-2.95^{**} / -2.48^{***}	0.027 / 0.008
R5: statspai/linearmodels LP-IV 交叉验证	3M	0.67	0.878

Table 4: 稳健性检验汇总 ($h = 10$, 每 10 亿美元对应的基点数) / Robustness summary ($h = 10$; bps per \$1 bn).

Two limitations deserve emphasis. First, our instrument is only moderately strong ($F = 6.1$); the LP-IV confidence bands are correspondingly wide, and we refrain from causal claims beyond what the OLS gradient, asymmetry and robustness patterns support jointly. Second, daily data cannot resolve intraday run dynamics; the event study mitigates but does not eliminate this constraint. Both limitations point toward future work with

intraday on-chain redemption data and richer instrument sets.

两点局限值得强调。第一，本文工具变量强度中等 ($F = 6.1$)，LP-IV 置信带相应较宽，因此本文不超出 OLS 梯度、非对称性与稳健性模式共同支持的因果主张。第二，日度数据无法刻画盘内挤兑动态；事件研究缓解但未消除这一约束。两点局限都指向未来工作：采用盘内链上赎回数据与更丰富的工具变量集。

8. Economic implications and policy discussion / 经济含义与政策讨论

The estimates quantify a channel that regulators have so far treated qualitatively. At the current scale of the sector—combined supply near \$263 bn, with typical five-day net flows of a few billion dollars—the baseline coefficients imply that a normal-sized \$3 bn weekly inflow compresses the 3-month bill yield by roughly 7 bps within two weeks. This is small relative to policy-rate moves but economically meaningful for bill spreads, repo pricing and the convenience premium, and it mechanically steepens the front of the curve: short yields fall while long yields are untouched.

本文的估计将一个此前仅被定性处理的渠道定量化。以行业当前规模（合计供应接近 2,630 亿美元，典型 5 日净流量为数十亿美元）计，基准系数意味着一笔常规规模的 30 亿美元单周净流入会在两周内将 3 个月期国库券收益率压低约 7 个基点。相对于政策利率变动这并不算大，但对国库券利差、回购定价与便利性溢价具有经济意义，并且会机械地使曲线前端变陡：短端下行而长端不受影响。

The asymmetry result reframes the financial-stability question. The binding margin is not expansion but redemption: outflows move yields roughly eight times more per dollar than inflows, and the effect is concentrated in exactly the states—crypto bear markets—in which redemptions cluster. Reserve mandates that push issuers into bills and repos improve token-holder safety but, by concentrating liquidation risk in the most systemically important money-market instruments, strengthen the coupling documented here. A natural policy complement is liquidity-aware reserve design: minimum cash and RRP shares that absorb redemptions without bill fire sales, calibrated precisely to the outflow coefficients we estimate.

非对称结果重新框定了金融稳定问题。约束性边际不是扩张而是赎回：流出对收益率的单位美元影响约为流入的八倍，且效应恰好集中在赎回易发的状态——加密熊市。将发行方推向国库券与回购的储备监管虽提升了代币持有者的安全性，但通过把清算风险集中于最具系统重要性的货币市

场工具，强化了本文所记录的耦合。一个自然的政策配套是流动性敏感的储备设计：设定最低现金与逆回购份额，使赎回无需抛售国库券即可吸收——其校准恰可依据本文估计的流出系数。

Finally, the weak-instrument finding carries a methodological message for this literature: at daily frequency, crypto-shock instruments explain only a modest share of stablecoin-flow variation, and point estimates should be reported alongside first-stage diagnostics and OLS-gradient cross-checks rather than in isolation.

最后，弱工具变量的发现对本文献具有方法论含义：在日度频率上，加密冲击类工具变量只能解释稳定币流量变异的一小部分；点估计应与第一阶段诊断和 OLS 梯度交叉验证一并报告，而不应孤立呈现。

9. Conclusion / 结论

This paper estimates the price impact of dollar-stablecoin flows on the U.S. Treasury yield curve with local projections, an Elastic-Net-based instrumental variable, interpretable machine learning, event studies and a VAR cross-check, using a newly assembled daily dataset for 2020–2026. Four facts emerge. Stablecoin inflows significantly compress 1- and 3-month yields (up to -8.5 and -6.4 bps per \$1 bn over 30 trading days) with no effect at 10–30 years; the effect is strongly asymmetric, with outflows dominating inflows by an order of magnitude and only outflows statistically significant; bear-market states amplify the response roughly tenfold; and the machine-learning instrument, while economically sensible and statistically relevant ($F = 6.1$), is not strong enough at daily frequency to support standalone causal claims—a caveat we regard as part of the results. For policy, the message is twofold: stablecoin issuers are now measurable marginal price-setters in the bill market, and reserve regulation should be designed around redemption-driven fire-sale risk rather than around expansion.

本文基于一套新整合的 2020–2026 年日度数据集，结合局部投影、基于 Elastic Net 的工具变量、可解释机器学习、事件研究与 VAR 交叉验证，估计了美元稳定币资金流对美国国债收益率曲线的价格影响。四个事实浮现：稳定币流入显著压缩 1 个月与 3 个月期收益率（每 10 亿美元在 30 个交易日内最多下行 8.5 与 6.4 个基点），而对 10–30 年期无影响；效应呈强非对称，流出主导流入约一个数量级，且仅流出统计显著；熊市状态将响应放大约十倍；机器学习工具变量虽在经济意义上合理且统计上相关（ $F = 6.1$ ），但在日度频率上强度不足以单独支撑因果主张——本文将这一警示视为结果的一部分。对政策而言，含义有两层：稳定币发行方现已成为国库券市

场可测度的边际定价者；储备监管应围绕赎回驱动的火烧售卖风险而非扩张来设计。

Declaration of competing interest / 利益冲突声明

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

作者声明不存在任何可能影响本文工作的已知经济利益冲突或个人关系。

Data availability / 数据可得性

All data and the fully reproducible notebook (`stablecoin_bonds_impact.ipynb`) are available in the project repository. FRED series are public [9]; stablecoin supply and on-chain series are from public blockchain data providers.

全部数据与完全可复现的 notebook (`stablecoin_bonds_impact.ipynb`) 见项目仓库。FRED 序列为公开数据 [9]；稳定币供应量与链上序列来自公开的区块链数据提供商。

References

- [1] Ahmed, R., Aldasoro, I., 2025. Stablecoins and Safe Asset Prices. BIS Working Paper No. 1270. Bank for International Settlements.
- [2] Cerutti, E., Firat, M., Hengge, M., Sagawa, T., 2026. Stablecoin Shocks. IMF Working Paper WP/26/44. International Monetary Fund.
- [3] Griffin, J.M., Shams, A., 2020. Is bitcoin really untethered? Journal of Finance 75, 1913–1964.
- [4] Jaswicz, S., 2025. Stablecoins Could Increase Treasury Demand, but Only by Reducing Demand for Other Assets. Technical Report. Federal Reserve Bank of Kansas City.
- [5] Jordà, Ò., 2005. Estimation and inference of impulse responses by local projections. American Economic Review 95, 161–182.
- [6] Krishnamurthy, A., Vissing-Jorgensen, A., 2012. The aggregate demand for treasury debt. Journal of Political Economy 120, 233–294.

- [7] Lundberg, S.M., Lee, S.I., 2017. A unified approach to interpreting model predictions, in: Advances in Neural Information Processing Systems 30 (NeurIPS), pp. 4765–4774.
- [8] Rigobon, R., 2003. Identification through heteroskedasticity. *Review of Economics and Statistics* 85, 777–792.
- [9] of St. Louis, F.R.B., 2026. Federal reserve economic data (fred). <https://fred.stlouisfed.org>.
- [10] Vayanos, D., Vila, J.L., 2021. A preferred-habitat model of the term structure of interest rates. *Econometrica* 89, 77–112.